

CLASS DESIGN DOCUMENT

Team Members

1. Abigail Johnson Inyang
2. Chinelo Nnamdi Kanu
3. Providence Inyange
4. Moses Kolaan

Selected Project: Bus Route & Ticketing Management System

Class 1

Class Name: Route

Purpose: Represents one bus route and stores its origin, destination, distance and base fare.

Attributes

Attribute	Data Type	Description
route_id	int	Unique ID for the route.
origin	str	Starting location.
destination	str	Destination location.
distance_km	float	Distance of the route in kilometres.
base_fare	float	Normal fare for the route.

Methods

Method	Parameters	Purpose
__init__()	route_id, origin, destination, distance_km, base_fare	Creates a Route object and stores the route details.
__str__()	None	Returns the route details as readable text.

Class 2

Class Name: RouteManager

Purpose: Manages route records in the system.

Attributes

Attribute	Data Type	Description
routes	dict	Stores Route objects using route IDs as keys.

Methods

Method	Parameters	Purpose
<code>__init__()</code>	None	Creates an empty dictionary for routes.
<code>generate_route_id()</code>	None	Returns the next route ID.
<code>add_route()</code>	origin, destination, distance_km, base_fare	Validates the details, creates a route and stores it.
<code>display_routes()</code>	None	Displays all stored routes.
<code>search_routes()</code>	destination	Searches for routes by destination.
<code>update_route()</code>	route_id, origin, destination, distance_km, base_fare	Updates an existing route.
<code>delete_route()</code>	route_id, trips	Deletes a route if it exists and is not being used by a trip.

Class 3

Class Name: Trip

Purpose: Represents one scheduled bus journey on a route.

Attributes

Attribute	Data Type	Description
trip_id	int	Unique ID for the trip.
route_id	int	ID of the route used for the trip.
travel_date	date	Date of travel.
departure_time	str	Departure time in HH:MM format.
arrival_time	str	Arrival time in HH:MM format.
total_seats	int	Total number of seats for the trip.

Methods

Method	Parameters	Purpose
<code>__init__()</code>	trip_id, route_id, travel_date, departure_time, arrival_time, total_seats	Creates a Trip object and stores the trip details.
<code>__str__()</code>	None	Returns the trip details as readable text.

Class 4

Class Name: TripManager

Purpose: Manages trip records and seat availability.

Attributes

Attribute	Data Type	Description
trips	dict	Stores Trip objects using trip IDs as keys.

Methods

Method	Parameters	Purpose
__init__()	None	Creates an empty dictionary for trips.
generate_trip_id()	None	Returns the next trip ID.
add_trip()	route_id, travel_date, departure_time, arrival_time, total_seats	Creates and stores a trip.
display_trips()	None	Displays all stored trips.
search_trips()	route_id, travel_date	Searches for trips by route ID and/or travel date.
update_trip()	trip_id, travel_date, departure_time, arrival_time	Updates a trip's date or times.
cancel_trip()	trip_id, ticket_manager	Cancels a trip if it does not have active tickets.
is_seat_available()	trip_id, seat_number, ticket_manager	Checks if a seat is free.
view_seat_layout()	trip_id, ticket_manager	Shows which seats are open or occupied.

Class 5

Class Name: Passenger

Purpose: Represents one registered passenger.

Attributes

Attribute	Data Type	Description
passenger_id	int	Unique ID for the passenger.
name	str	Passenger name.
phone	str	Passenger phone number.

Methods

Method	Parameters	Purpose
__init__()	passenger_id, name, phone	Creates a Passenger object and stores the passenger details.

Method	Parameters	Purpose
__str__()	None	Returns the passenger details as readable text.

Class 6

Class Name: PassengerManager

Purpose: Manages passenger records in the system.

Attributes

Attribute	Data Type	Description
passengers	dict	Stores Passenger objects using passenger IDs as keys.

Methods

Method	Parameters	Purpose
__init__()	None	Creates an empty dictionary for passengers.
generate_passenger_id()	None	Returns the next passenger ID.
register_passenger()	name, phone	Validates and registers a passenger.
search_passenger()	passenger_id, name, phone	Searches for a passenger.
display_passengers()	None	Displays all registered passengers.
update_passenger()	passenger_id, name, phone	Updates a passenger's name or phone number.

Class 7

Class Name: BusPass

Purpose: Represents a bus pass issued to a passenger.

Attributes

Attribute	Data Type	Description
pass_id	int	Unique ID for the bus pass.
passenger_id	int	ID of the passenger who owns the pass.
pass_type	str	Type of pass: student, senior or priority.
issue_date	str	Issue date in YYYY-MM-DD format.

Attribute	Data Type	Description
expiry_date	str	Expiry date in YYYY-MM-DD format.
status	str	Current status of the pass.

Methods

Method	Parameters	Purpose
__init__()	pass_id, passenger_id, pass_type, issue_date, expiry_date, status	Creates a BusPass object and stores the pass details.
__str__()	None	Returns the bus pass details as readable text.

Class 8

Class Name: BusPassManager

Purpose: Manages bus pass records and checks whether passes can be used.

Attributes

Attribute	Data Type	Description
bus_passes	dict	Stores BusPass objects using pass IDs as keys.

Methods

Method	Parameters	Purpose
__init__()	None	Creates an empty dictionary for bus passes.
generate_pass_id()	None	Returns the next bus pass ID.
issue_pass()	passenger_id, pass_type, issue_date, expiry_date, passengers	Creates a pass for an existing passenger.
display_passes()	None	Displays all bus passes.
search_pass()	pass_id, passenger_id	Searches for bus passes.
is_pass_valid()	pass_id, travel_date, passenger_id	Checks the pass status, owner and valid date range.
renew_pass()	pass_id, new_expiry_date	Changes the expiry date and reactivates the pass.
set_pass_status()	pass_id, status	Changes the status of a bus pass.

Class 9

Class Name: Ticket

Purpose: Represents one ticket purchased for a passenger, trip and seat.

Attributes

Attribute	Data Type	Description
ticket_id	int	Unique ID for the ticket.
passenger_id	int	ID of the passenger.
trip_id	int	ID of the trip.
seat_number	int	Seat number on the trip.
pass_id	int or None	Bus pass ID used for the ticket, if any.
price	float	Final ticket price.
purchase_date	date	Date the ticket was purchased.
status	str	Ticket status: active or cancelled.

Methods

Method	Parameters	Purpose
__init__()	ticket_id, passenger_id, trip_id, seat_number, pass_id, price, purchase_date, status	Creates and validates a Ticket object.
_validate_id()	value, field_name	Checks that an ID is a positive integer.
_validate_seat_number()	seat_number	Checks that the seat number is valid.
_validate_price()	price	Checks that the ticket price is valid.
_validate_pass_id()	pass_id	Checks the optional bus pass ID.
_validate_status()	status	Checks that the ticket status is allowed.
cancel()	None	Changes an active ticket to cancelled.
is_active()	None	Returns whether the ticket is active.
__str__()	None	Returns the ticket details as readable text.

Class 10

Class Name: TicketManager

Purpose: Manages ticket purchases, fares, searches, cancellations and seat checks.

Attributes

Attribute	Data Type	Description
tickets	dict	Stores Ticket objects using ticket IDs as keys.

Methods

Method	Parameters	Purpose
<code>__init__()</code>	None	Creates an empty dictionary for tickets.
<code>add_ticket()</code>	<code>ticket</code>	Stores a Ticket object.
<code>generate_next_ticket_id()</code>	None	Returns the next ticket ID.
<code>calculate_ticket_price()</code>	<code>base_fare</code> , <code>pass_type</code>	Calculates the final ticket price.
<code>search_ticket()</code>	<code>ticket_id</code>	Searches for a ticket by ID.
<code>display_all_tickets()</code>	None	Displays all tickets.
<code>display_active_tickets()</code>	None	Displays active tickets.
<code>display_cancelled_tickets()</code>	None	Displays cancelled tickets.
<code>cancel_ticket()</code>	<code>ticket_id</code>	Cancels a ticket.
<code>get_tickets_by_passenger()</code>	<code>passenger_id</code>	Returns tickets for one passenger.
<code>get_tickets_by_trip()</code>	<code>trip_id</code>	Returns tickets for one trip.
<code>seat_is_taken()</code>	<code>trip_id</code> , <code>seat_number</code>	Checks whether a seat already has an active ticket.
<code>count_active_tickets()</code>	<code>trip_id</code>	Counts active tickets for a trip.
<code>purchase_ticket()</code>	<code>passenger_id</code> , <code>trip_id</code> , <code>seat_number</code> , <code>routes</code> , <code>passengers</code> , <code>trips</code> , <code>passes</code> , <code>pass_id</code>	Checks the booking details, calculates the fare and creates the ticket.